

1 Interesting Concepts related to Vision

1.1 Why Every Rental Property looks Better than Reality

Rental sites generally take a bunch of images of the house and then they turn the saturation way up to make an image appear much more vivid and vibrant. This is a very common trick used to make properties look more attractive when selling houses.

1.2 Our Eyes Never Remains Still

Three invisible movements keeps our eyes alive. These movements exist to prevent vision from fading, otherwise our eyes will lose sensitivity if they stare at the same point too long.

1.3 Our Eyes Are Shaped by Forests

We see more shades of green than any other color, because cones in our eyes respond mainly to wavelengths close to green and extend to red. We have many more red and green cones than blue ones. We evolved this ability to spot hunting targets and dangers in forests.

1.4 Octopuses Don't Have a Blind Spot - But Humans Do

We have a blind spot because our optic nerve comes out of the retina into the eye and then back out. Octopuses, which have very similar eyes, have no blind spot because their optic nerves exit in the opposite direction. Light cannot pass through nerves, which is why our blind spot exists.

1.5 Generating Fake Training Data for Training Models

By converting an image to HSV color space and modifying the hue channel, we can generate new dataset for training machine learning models. These hue-shifted images look very similar to humans but appear different to computers, allowing researchers to significantly increase dataset sizes without new annotations.

2 Types of Healthcare Reports

Healthcare reports are formal documents that record, communicate, and analyze information across every dimension of patient care and hospital operations. They broadly fall into five categories:

2.1 Clinical Reports

Clinical Reports form the core of patient care documentation. These include History and Physical Examinations, Progress Notes, Operative Reports, Physician Orders, Discharge Summaries, and Nursing Notes. They enable longitudinal tracking of a patient's health journey and support informed clinical decision-making.

2.2 Diagnostic Reports

Diagnostic Reports provide objective, measurable data for diagnosis. These include Laboratory Reports, Pathology Reports, Radiography Reports, EKG Reports, Physical Therapy Reports, and Dietary Reports. Medical imaging reports — radiology, MRI, CT scans, and ultrasounds — fall under this category and are among the most complex and consequential diagnostic documents produced in healthcare.

2.3 Administrative Reports

Administrative Reports support the operational management of healthcare institutions. Administrative data is generated during every interaction with the healthcare system — a doctor's visit, a diagnostic procedure, hospital admission, or filling a prescription. This data is gathered for administrative or billing reasons but can also be used to analyze healthcare delivery, advantages, disadvantages, and cost-effectiveness.

2.4 Financial Reports

Financial Reports translate clinical services into billing information. Analysis of financial data allows healthcare practitioners to assess how resources are used within the organization, evaluate expenses, staff allocation, and equipment costs, and identify areas of inefficiency or overspending to improve overall cost-effectiveness.

2.5 Regulatory and Public Health Reports

Regulatory and Public Health Reports are mandatory external submissions. Commonly used administrative aggregate reports include basic healthcare statistical reports, claims denial reports, and cost reports. These are often submitted to government bodies, insurers, and public health authorities to ensure quality compliance and patient safety standards are met

3 Medical Imaging Reports

A medical imaging report is the formal document produced by a radiologist after interpreting a diagnostic scan — such as an X-ray, CT scan, MRI, ultrasound, or nuclear medicine study. It is the primary channel of communication between the radiologist and the treating clinician, translating complex visual data into clinically actionable language.

The radiology report also carries significant legal weight — thorough documentation can be the main differentiator between radiologic errors and negligent care constituting malpractice, as courts and juries tend to assume that the absence of documentation implies that the contested facts or events did not occur.